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Q1.(String Indexing)

```
text = input("Enter a string: ")

print("First character:", text[0])

print("Last character:", text[-1])

print("3rd character from the start:", text[2])

print("2nd character from the end:", text[-2])
```

Q2.(String Slicing)

```
text = input("Enter a string: ")

print("First 5 characters:", text[:5])

print("Last 4 characters:", text[-4:])

print("Reversed string:", text[::-1])

print("Every 2nd character:", text[::2])
```

Q3.(String Membership,)

```
main_string = input("Enter the main string: ")

substring = input("Enter the substring: ")

if substring in main_string:

    print("Substring found!")

if substring not in main_string:

    print("Substring not found!")
```

Q4.(len() Function)

```
text = input("Enter a string: ")

length = len(text)

print("Length of the string:", length)

print("Last character:", text[length - 1])

if length % 2 != 0:

    print("Middle character:", text[length // 2])

else:

    print("The string has an even length, so there is no single middle character.")
```

Q5.(range() - Basic Forms)

```
print("Numbers from 0 to 10:")

for i in range(11):

    print(i)

print("\nNumbers from 5 to 15:")

for i in range(5, 16):

    print(i)

print("\nOdd numbers from 1 to 21:")

for i in range(1, 22, 2):

    print(i)
```

Q6.(range() with Negative Step)

```
print("Numbers from 20 down to 10:")
for i in range(20, 9, -1):
    print(i)
print("\nNumbers from 15 down to 1 in steps of 2:")
for i in range(15, 0, -2):
    print(i)
```

Q7.(Combined: Strings + range()

```
text = input("Enter a string: ")
print("Characters with their index:")
for i in range(len(text)):
    print("Index", i, ":", text[i])
print("\nString in reverse order:")
for i in range(len(text) - 1, -1, -1):
    print(text[i], end="")
```

Q8.(Membership with range())

```
num = int(input("Enter a number: "))
if num in range(1, 51):
    print(num, "is present in range(1, 51).")
else:
    print(num, "is NOT present in range(1, 51).")
```

```
if num in range(10, 100, 5):
    print(num, "is present in range(10, 100, 5).")
else:
    print(num, "is NOT present in range(10, 100, 5).")
```

Q9.(Slicing + range()

```
print("First 10 even numbers (2 to 20):")
for i in range(2, 21, 2):
    print(i)
print("\nNumbers from 1 to 30 in steps of 3:")
for i in range(1, 31, 3):
    print(i)
text = "PythonProgramming"
print("\nOriginal String:", text)
print("Python:", text[:6])
print("Programming:", text[6:])
print("Reverse:", text[::-1])
```

Q10.(Mini Project - Combined)

```
user_string = input("Enter a string: ")
length = len(user_string)
print("\nLength of the string:", length)
middle = length // 2
```

```
print("First half :", user_string[:middle])
print("Second half:", user_string[middle:])
if "python" in user_string.lower():
    print("The word 'python' is present.")
else:
    print("The word 'python' is not present.")
print("\nCharacters with Positive and Negative Indices:")
for index in range(length):
    negative_index = index - length
    print("Positive Index:", index,
          "Negative Index:", negative_index,
          "Character:", user_string[index])
print("\nString in Reverse:", user_string[::-1])
```